Who could reach whom: modelling radio and phone coverage on the CFB Wainwright ranges, with the weather and the fire

Kevin J.S. Duska Jr. · Oct. 29, 2026 · v6 working material

When the wind turned on the afternoon of Oct. 17, the soldiers on the pairs range depended on radios, and possibly phones, to hear a warning and to call for help. DND hasn't said what equipment they had or whether any link failed. So I modelled what the terrain around the burn allows: how far a walkie-talkie, a vehicle radio, a base station and a phone could reach. I tested that under the actual weather of that day, and with the fire itself in the way.

> **Read this as a model, not a finding of fact.** I don't know which radios DND used, on which frequencies, or where Range Control's antennas are. I've used generic equipment of the kind found on Canadian ranges and set out every assumption below, so readers can see how much each one moves the result. The model shows what was physically likely. It can't show what happened.

1. Summary

- **Walkie-talkies were short-range on this ground.** A 5 W handheld talking to another handheld works reliably only within about **1 to 1.5 km** of the pairs range (0.9 to 1.9 km across my sensitivity tests). Two low spots inside the burn, about 2.4 km west and south of the pairs range, were essentially out of handheld reach (3–5% chance).
- **A vehicle radio changes everything locally.** A 50 W vehicle set at the pairs range can be heard by handhelds out to about **4 km**, and by other vehicle sets to about **8 km**. A handheld's reply back to the vehicle carries less far, about 2 to 3 km.
- **Reaching a base station depends on where it is and how high its antenna sits.** For a station assumed in the garrison area about 6 km north, with a 20 m mast, a handheld at the pairs range gets through about **75%** of the time. A 10 m mast would drop that to about 42%; a 40 m mast would raise it to about 95%.
- **The town's fire radio probably reached the pairs range.** Wainwright Fire's licensed VHF system would connect a firefighter's portable at the pairs range about **75%** of the time, if its antenna is about 30 m up.
- **Phones:** Rogers about **87%** at the pairs range; TELUS and Bell about **33%**. A 911 call can use either network.
- **The day's weather didn't matter.** I measured how strongly the atmosphere bent radio waves on Oct. 17, using that day's weather-balloon soundings. Rerunning the model with those values moved results by about 1 to 4 percentage points.
- **The fire itself may have.** Flames and the hot, turbulent plume can absorb and scatter radio signals. If they added just 6 dB of loss inside the burn, the handheld-to-base link from the pairs range would fall from about 75% to about 50%, Rogers phone signal from 87% to 73%, and TELUS from 33% to 19%. The 6 dB figure is my assumption, not a measurement. The direction of the effect is not in doubt.

2. Why this matters

Maj.-Gen. Wade Rutland has said range control had known about a fire about a kilometre away for "several hours", and that it was "not originally assessed as a threat" (as reported by The Globe and Mail). Environment Canada's wind warning covered Wainwright from 2:14 p.m.; the wind turned at about 3:16 p.m.; DND says the

fire reached the soldiers "just before 4 p.m." Under the base's Range Standing Orders, a forecast of adverse weather is to be passed to all concerned "immediately". Whether it could have been, and whether the troops could call for help once the fire turned, comes down to who could reach whom by radio. This report sets out what the ground and the equipment allowed. It doesn't say what was said.

3. What I modelled

Link	Band	Transmit power	Antennas	Receiver threshold	Loss the link can tolerate
A. Handheld ↔ handheld	Low VHF, 50 MHz	5 W (37 dBm)	−5 dBi each, 1.5 m	−113 dBm	140 dB
B. Vehicle → handheld	Low VHF, 50 MHz	50 W (47 dBm)	Vehicle 0 dBi at 3 m; handheld −5 dBi at 1.5 m	−113 dBm	155 dB (reply: 145 dB)
C. Vehicle ↔ vehicle	Low VHF, 50 MHz	50 W	0 dBi each, 3 m	−113 dBm	160 dB
D. Handheld ↔ base station (assumed, garrison area)	Low VHF, 50 MHz	Handheld 5 W; base 50 W	Base 3 dBi at 20 m	−113 dBm	148 dB (handheld's reply limits it)
E. Wainwright fire base ↔ portable	VHF, 169 MHz (licensed)	Portable 5 W	Base 6 dBi at an assumed 30 m; portable −2 dBi	−116 dBm	157 dB
F. Phone (from the reception report)	LTE, 700 MHz	Licensed Rogers and TELUS towers, ISED heights; phone outdoors at 1.5 m			140 dB

Low VHF (30–88 MHz) is the traditional band for army combat radio. Range and fire-service nets in Alberta commonly use high VHF or UHF, so I also tested the handheld case at 150 MHz and 400 MHz. Link E uses the Town of Wainwright fire department's frequencies and site from ISED's licence data. The base-station location in link D is my assumption: Range Control's antenna site isn't public.

4. Method

Terrain and grid

I used the Copernicus GLO-30 digital surface model, which includes trees and buildings, reprojected to a 30 m grid (UTM zone 12N). Results are computed on a 60 m grid covering about 24 by 23 km around the pairs range, and smoothed over 180 m for the maps.

Signal loss along each path

For every grid point, I traced the terrain profile from the transmitter at 360 sample points, adjusted for Earth's curvature and atmospheric bending (the "k-factor"). Total loss is:

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Loss = max(free-space loss, plane-earth loss) + terrain diffraction + clutter allowance
free-space = 32.45 + 20·log10(f MHz) + 20·log10(d km)
plane-earth = 40·log10(d m) − 20·log10(h_tx m) − 20·log10(h_rx m)
diffraction: knife-edge loss at the main obstruction, plus half the loss at the worst
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obstruction on each side of it (simplified Deygout method)
clutter: 6 dB for radios, 8 dB for phones (vegetation, body, vehicles)
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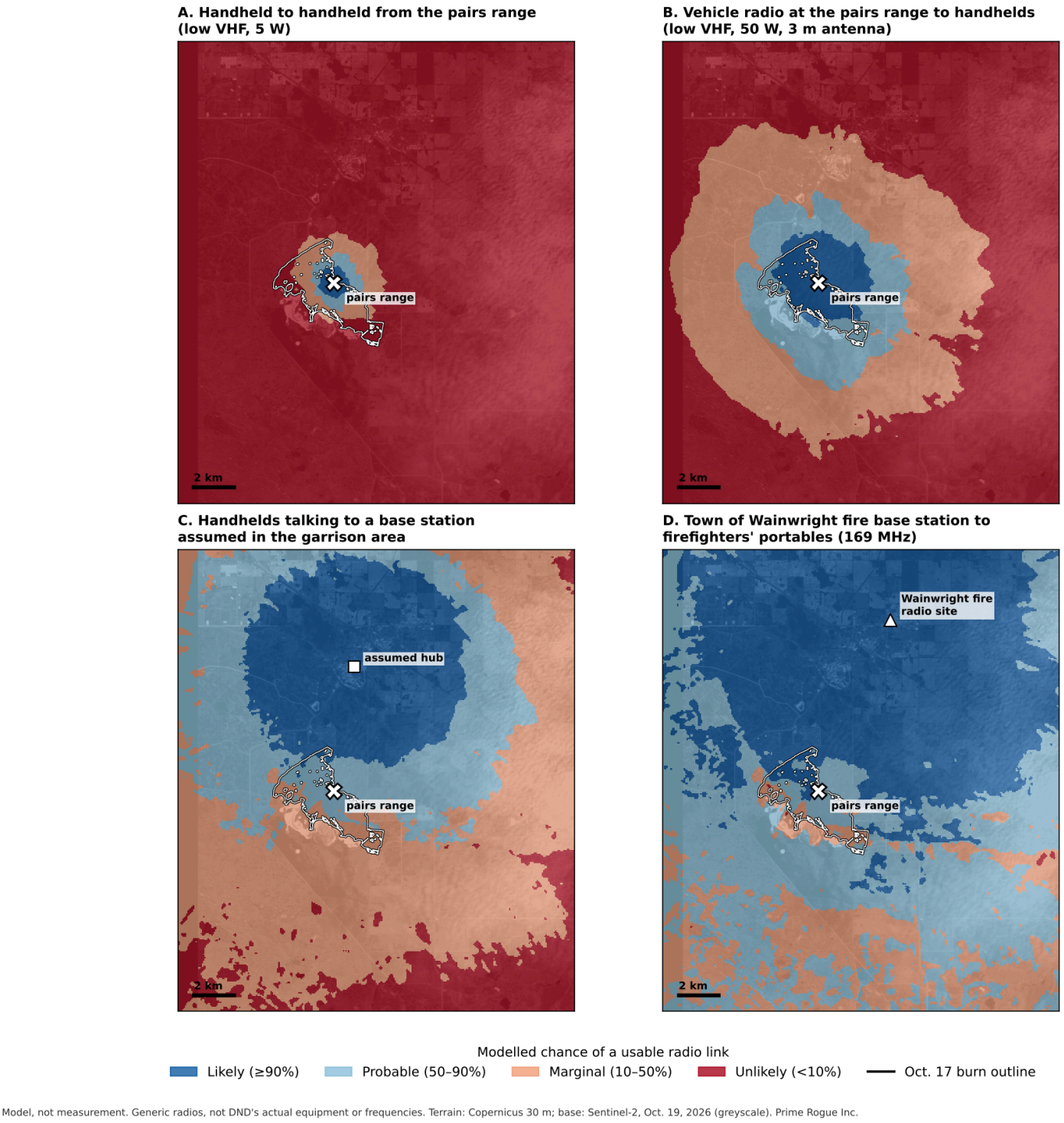
From loss to probability

A link counts as usable when total loss is below what the equipment tolerates (section 3). Real signal levels vary from place to place because of small terrain features, vegetation and how a radio is held. I modelled that variation as 8 dB, so each point gets a probability: likely (90% or more), probable (50–90%), marginal (10–50%) or unlikely (under 10%).

Points I report

Values at named places are the median over 300 m around each one, reflecting how uncertain the locations are. The places are: our best estimate of the pairs range (52.7695 N, 110.9005 W); DND's official grids for Range 2 and Range 15; and two low spots inside the burn, about 2.4 km to the west (the "west basin") and south (the "south lakes").

5. Results



Model, not measurement. Generic radios, not DND's actual equipment or frequencies. Terrain: Copernicus 30 m; base: Sentinel-2, Oct. 19, 2026 (greyscale). Prime Rogue Inc.

Link (baseline)	Share of burned area at ≥50%	Reach (90% of good points within)	Pairs range	West basin	South lakes
A. Handheld ↔ handheld, from the pairs range	31%	1.3 km	n/a (origin)	3%	5%
B. Vehicle at the pairs range → handhelds	100%	4.3 km	n/a (origin)	77%	86%

C. Vehicle ↔ vehicle	100%	7.8 km	n/a (origin)	98%	99%
D. Handheld ↔ assumed garrison base (20 m mast)	60%	11.3 km	75%	35%	35%
E. Wainwright fire base ↔ portable (169 MHz)	75%	12.5 km	75%	47%	47%
F. Phone, Rogers	79%	n/a	87%	see the reception report	
F. Phone, TELUS / Bell	39%	n/a	33%	see the reception report	

The weak areas are the same in every model: the low basin west of the pairs range and the lakes along the burn's south side. That's where terrain blocks the paths to every likely transmitter. Someone in either spot would have had a hard time on a handheld, and only a fair chance on a phone.

6. Weather and the fire

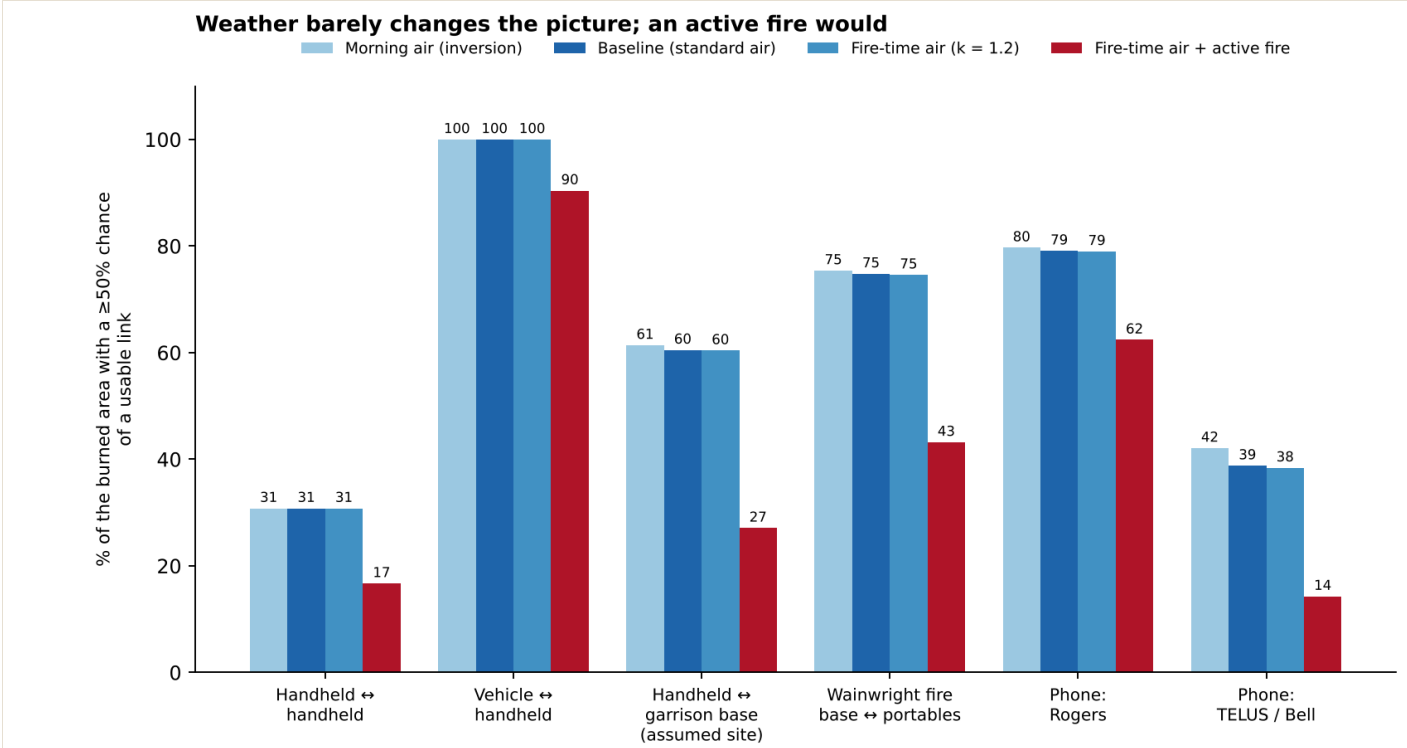
What the weather data shows

At these frequencies, rain, wind and ordinary smoke have almost no effect on signal strength. The weather factor that does matter is how the atmosphere bends radio waves, which depends on how temperature and humidity change with height. I calculated it from Environment Canada's weather-balloon soundings at Stony Plain on Oct. 17. Stony Plain is near Edmonton, about 200 km west, and it's the nearest balloon site.

- **6 a.m.:** a strong surface inversion, 8 °C at the ground and 19 °C about 230 m up, with very dry air above. Refractivity fell by about 176 units per km in the lowest 100 m, enough to trap radio waves near the ground and extend range well beyond normal.
- **6 p.m., behind the front:** well-mixed air, with refractivity falling about 26 units per km in the lowest 300 m. That's a k-factor of about 1.2, slightly less bending than the standard 1.33. The fire reached the soldiers in this post-front air.

What the active fire may add

Flames contain hot, partly ionized gases, and a wind-driven fire throws up turbulent hot air and dense smoke. These can absorb and scatter radio signals passing through or near them. I found no measurement for a grass fire like this one, so I tested an assumed extra 6 dB of loss for any point inside the burn. That's roughly the effect of cutting transmit power to a quarter.



Model, not measurement. Refraction (k) from ECCC Stony Plain soundings, Oct. 17 6 a.m. (inversion) and 6 p.m. (post-front). Active-fire case assumes 6 dB extra loss inside the burn. Prime Rogue Inc.

At the pairs range (300 m median)	Morning air	Baseline	Fire-time air	Fire-time air + active fire
Handheld ↔ assumed garrison base	75%	75%	75%	49%
Wainwright fire base ↔ portable	76%	75%	75%	55%
Phone, Rogers	87%	87%	87%	73%
Phone, TELUS / Bell	36%	33%	32%	19%

Bottom line: the weather that day doesn't explain any communication problem; the bending effect is too small over distances of a few kilometres. If radios or phones struggled, the more likely physical causes are the terrain (the low spots) and the fire itself, plus the human and equipment factors this model can't see.

7. How sensitive the results are

Change	Effect
Handheld link 6 dB worse or better (antenna held low, worn battery, better radio)	Handheld reach 0.9 km to 1.9 km; west basin 0–12%
Handheld at 150 MHz or 400 MHz instead of 50 MHz	Reach 1.2 km and 1.1 km. Higher bands are blocked more by this terrain.
Vehicle link 6 dB worse or better	Vehicle → handheld reach 2.9 km to 5.8 km
Base station mast 10 m / 20 m / 40 m	Pairs range 42% / 75% / 95%
Base station 2 km closer / 2 km farther	Pairs range 93% / 50%
Wainwright fire antenna 15 m instead of 30 m	Pairs range 40% instead of 75%
Atmospheric bending (k = 1.2 to 4)	About 1 to 4 percentage points

Active fire adds 6 dB	See section 6: the largest single effect after equipment and antenna height

8. Phones and radios compared

Channel	At the pairs range	In the low spots	Comment
Handheld to handheld	Fine within about 1 km	Out of reach from the pairs range	The usual squad-level channel; short range on this ground
Vehicle radio	Strong	Good (77–99%)	The strongest local link, if a vehicle set was there and manned
Base station (assumed)	Probable (75%)	Marginal (35%)	Depends heavily on the real site and mast
Phone, Rogers	Probable to likely (87%)	Weaker	Also carries 911 calls from any phone
Phone, TELUS / Bell	Doubtful (33%)	Poor	Can still reach 911 through Rogers

9. What it means, and what I can't say

- On this terrain, a range relying mainly on handhelds would have had real gaps beyond about a kilometre. Vehicle sets and a well-sited base station close most of them.
- The weather on Oct. 17 didn't degrade radio or phone signals in any meaningful way. The fire may have, in the area that mattered most at the moment it mattered most.
- I can't say what equipment was on the pairs range, whether nets were busy, whether radio checks were done, or whether any call or warning failed to get through. Those are records DND holds.

Questions for DND

1. What radios and nets were the units on the pairs range using: handhelds, vehicle sets, or a relay on high ground? On which bands?
2. Where are Range Control's antennas, and how high?
3. Were radio checks logged on Oct. 17? Was any loss of communication reported before or during the fire?
4. Were personal phones carried on the range, and was the first emergency call made by radio or by phone?

The records behind questions 1 to 3 are covered in my access-to-information request (categories A–D). Question 4 has been put to Emergency Health Services Alberta.

10. Limits

These are generic radios, not DND's equipment; real tactical radios may be digital, frequency-hopping or relayed, and some outperform these assumptions. Antenna patterns, directional antennas and network loading aren't modelled. The surface model counts tree canopy as an obstacle, which can overstate losses in wooded patches and understate them on burned ground after the fire. The 8 dB variation and the receiver thresholds are typical values, not measured ones. The soundings are from 200 km away. The fire-loss value is an assumption chosen to show direction and scale. Location of the pairs range is approximate (±300 m).

11. Files and reproducibility

The scripts, inputs and outputs are in the v6 folder of the public archive (`data/v6/comms_reception/` in github.com/primerogueinc/cfb-wainwright-fire-archive-2). They include the tower list, model and sensitivity scripts, the weather-scenario runs (JSON) and every figure. The terrain model and Sentinel-2 imagery are open data and are downloaded by the scripts.

Sources

- ISED, Spectrum Management System data: radio and spectrum licence extracts, Oct. 5, 2026
- Copernicus DEM GLO-30 (ESA / Copernicus)
- Sentinel-2 L2A, Oct. 19, 2026 (Copernicus, via Earth Search)
- ECCC MSC Datamart soundings, Stony Plain (CWSE): 6 a.m. Oct. 17, 6 p.m. Oct. 17
- ECCC wind warning, 2:14 p.m. Oct. 17
- WiGLE.net cell sightings (used as a check in the reception report)
- DND, Oct. 18 news release; CFB Wainwright Range Standing Orders (2024), paras 214–216

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